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OLCF HPC Operations



Hewlett Packard
Enterprise



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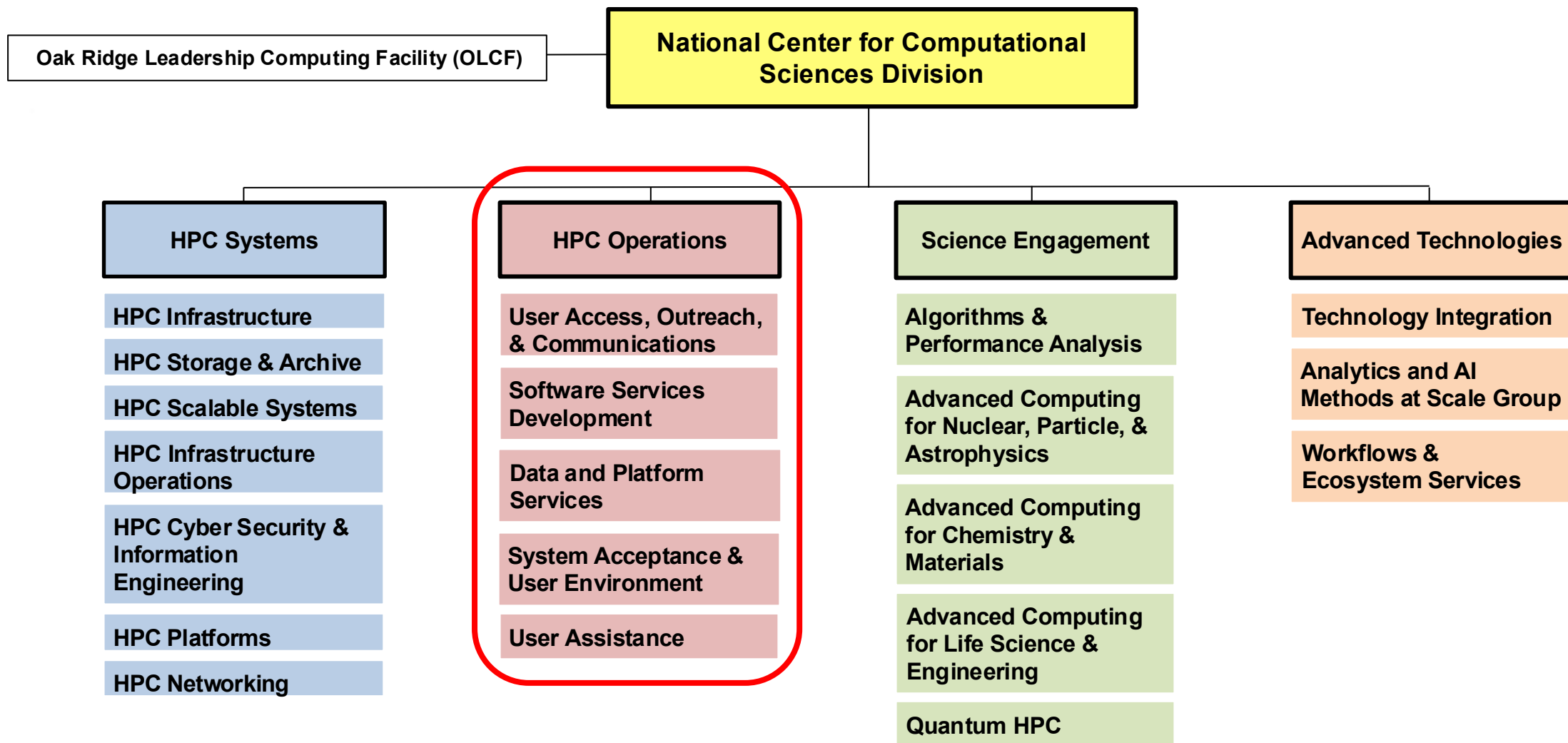


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The HPC Operations Section

- Handles the day-to-day operations of our production resources (or systems on their way to production)
 - **User Access, Outreach, and Communications***
 - Onboard and offboard all users across all programs within NCCS
 - **Software Services Development***
 - Designs and builds custom software to support users and projects (API automation)
 - **Data and Platform Services**
 - Provide services for data ingestion, management, and publication
 - Develop tools that interface with our Kubernetes-based services like Jupyter and Slate
 - **System Acceptance and User Environment**
 - Provision the software environment on HPC systems
 - Acceptance and mid-life testing of HPC systems
 - **User Assistance***
 - Documentation, training, debugging of bleeding-edge workloads on production HPC systems

User Access, Outreach, and Communications (Accounts)

Concerns that apply both to HPC and QC

- Volume of user applications and project proposals
- Restricted users?
 - E.g., IBM member agreements
 - Citizenship restrictions / export control (also related to scale of system?)
- What are we allowed to publicly share about the machine?

Project proposal differences for QC

- Uniqueness of how time is allocated?
- New allocation program?

Software Services Development (SSD)

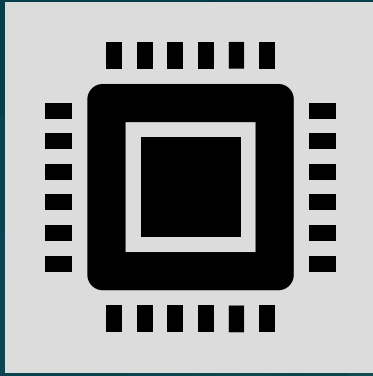
SSD stands up our Resource and Allocation Tracking System (RATS)

- Tracks OLCF projects, allocations, users, filesystem quotas
- Creates unix groups, users, and directories on HPC systems
- Syncs with SLURM to impose partition limits

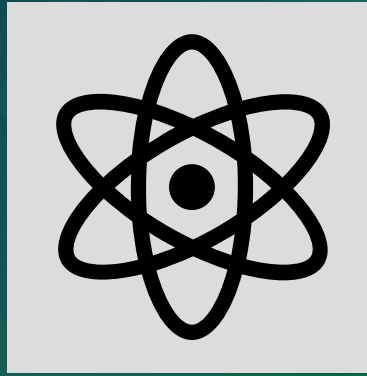
What the SSD team is concerned with

- Establishing the **allocation units** we want to charge
 - QPU (“node”) hours? Qubit (“core”) hours?
 - Policy on allocation penalties
- All QCUP projects are currently “Open” **security enclave**
- RATS controls SLURM **scheduler configurations** that enforce limits on projects / partitions
 - Introduce QPU as a Slurm partition?
 - Do we couple the allocations or is it a separate allocatable resource? Gets problematic if just a partition
- Is **API behavior** like “on-prem” cloud (e.g., depends on GUI interface still) or “real” on-prem where nothing is user facing
- **Reporting capabilities?**
 - Seems like a “nice to have”, but DOE will care about this

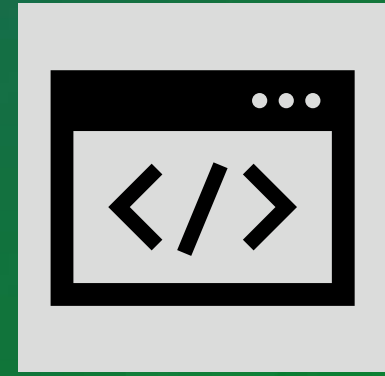
Data and Platform Services (DAPS)



- With commercial quantum hardware using cloud-style provisioning, how would this map on to the current kubernetes cluster model
- SLURM plugins are around, but what about kubernetes support?



- What quantum simulators/SDKs make sense to provide through Jupyter or Kubernetes to help on-board future researchers
- How do we get all the SDKs to play nicely in a Jupyter kernel (e.g., would various separate kernels be better without things getting too messy / overwhelming?)



- Can web portals or GUI services utilize OLCF's Slate service?

System Acceptance & User Environment (SAUE)

Verifiability and Correctness

- In traditional FP64 HPC, we check bit-wise reproducibility repeatedly, and that's how we know a GPU, CPU, & memory is working
- Due to statistical nature of QC, what is our metric to verify “correct” behavior
- How do we know when we have sampled for long enough that isn't purely application-specific?

Hardware-dependent Benchmarks and Variability

- Need to establish a consistent benchmark as a community
- Application behavior is strongly dependent on hardware implemented (see QCUF talk that compared different quantum backends to an RNG)

Clarity on “Acceptance”

- Have to be careful about overloading this term
- “vendor acceptance” versus “science engagement acceptance” versus “full acceptance”

Software

- Proprietary software and licenses (e.g., Quantum Brilliance emulator / noise model)
- Not feasible to make a user accept a license agreement by doing a `module load`
 - In a testbed env, misc. users could have `root` access which could bypass loading restrictions
- Highly prefer open-source

User Assistance (UA)

UA is the “glue”

- Overlaps all the other groups in Operations, so has all the same concerns
- Would need to be involved in full integration process since we’re expected to support the full process
- Would need a view into calibration to temper user expectations and to inform users

Best Practices

- Need to pass along lessons-learned about training, documentation, and support from HPC world
- Allocation policies, file management, notification behavior, change management procedures, etc.

Ticket Escalation

- Need to have a pathway for technical user questions that are highly quantum (similar to QCUP model)
 - Or more expertise is needed internally on UA
- Not new, have vendor ticket pipeline and the “Center of Excellence” model in the HPC world

Notifications & Communication

- Uptime and downtime concerns
- What is considered an “outage” relative to rest of HPC center
- SLURM reservations for outages?
- Impact on non-quantum users (e.g., if not doing an isolated set of nodes, then outages could impact all HPC users?)
- Health status visibility and communication is complex (similar to filesystem health status “volatility”)

Misc. Questions

- Where/when does the specialized QC/HPC integration line shift toward production efforts in general?
 - Don't want to reinvent the wheel – keeping us (Ops.) in the loop is desired even though still in testbed stage
- What warrants leadership-scale HPC resources (e.g., why connect QPU to Frontier instead of Andes)
- How does HPC node sharing (not QPU sharing) affect all of this?
 - Non-exclusive HPC nodes likely to be more common in the future
- Security (e.g., for cloud-like behavior, storing API tokens would raise flags)
- Networking (firewall / port expectations?)
- How will it be connected to the HPC parallel filesystem (e.g., lustre / GPFS) and/or NFS?